

Senior Research Engineer
elefthei@mit.edu
+1 (267) 968-3532

Eleftherios (Lef) IOANNIDIS

elefthei.github.io
github.com/elefthei
linkedin.com/in/elefthei

EDUCATION

PHD IN COMPUTER SCIENCE, *University of Pennsylvania*, Philadelphia, PA

Thesis: “*Correct Programs, Executed Correctly: Verifying Specifications and Executions*” (2025).

Advisors: Sebastian ANGEL & Steve ZDANCEWIC

MENG IN COMPUTER SCIENCE, *Massachusetts Institute of Technology* (MIT), Cambridge, MA

Thesis: “*Extracting and optimizing low-level bytecode from high-level verified Coq*” (2019).

Advisors: Nickolai ZELDOVICH & Frans KAASHOEK

👤 **Adoption:** Rocq-to-C++ extraction adopted by Bloomberg (bloomberg/crane) in 2025.

BSC IN COMPUTER SCIENCE, *Massachusetts Institute of Technology* (MIT), Cambridge, MA

Thesis: “*Parallel annotations for polyhedral optimizations in LLVM*” (2015).

Advisor: Saman AMARASINGHE

🏆 **SuperUROP:** Advanced Undergraduate Research in EECS Award (2015).

EMPLOYMENT

2025 - PRESENT Senior Research Software Engineer, Microsoft Research, Redmond, WA

🧠 Working in the RiSE group, focusing on LLM assisted program verification and proof synthesis.

✅ AI-assisted **formal verification** of systems using **F***, **Pulse** and **Verus** proof assistants.

🔍 AI-assisted **bug finding** at scale in the Core **Windows OS**.

SUMMER 2022 Research Scientist Intern, AMAZON, Automated Reasoning Group, Arlington, VA

✅ Worked in the **formalization** of the Cedar authorization language and the Cedar validator.

🔗 Implemented a novel **type inference** algorithm for Cedar including singleton and capability types.

2018 - 2019 Investment Engineer, BRIDGEWATER Associates, Westport, CT

📊 Created **APIs** used for **big-data** quantitative research, analytics and **visualization**.

📈 Implemented complex risk-controls and hedging **algorithms** used daily by Trade Generation.

👤 Taught the **Scala** and **SQL** programming languages to more than 100 traders and engineers.

2016 - 2018 Founding Engineer & Architect, UNIFYID (acquired by PROVE), San Francisco, CA

🏠 Designed and implemented a **microservice** based **back-end** on **AWS** (20 services).

🧠 Implemented a real-time **Machine Learning** service, for high-throughput inference (3000 req/sec).

🔒 Developed certificate management systems and implemented **end-to-end encryption**.

💰 **Series-A:** \$23m raised on a \$100m valuation, led by a16z, NEA, StartX and more.

🏆 **1st place**, RSA Innovation Sandbox 2018 Startup Award (Boston, MA).

🏆 **1st place**, SXSW 2018 Startup Competition Award (Austin, TX).

🏆 **Runner-up**, TechCrunch Disrupt 2017 Startup Award (San Francisco, CA).

2015 - 2016 Software Security Engineer, APPLE, Cupertino, CA

🔗 Contributed to the **LLVM compiler**, focus on compiler optimizations for program security.

🔒 Implemented **cryptographic algorithms** for end-to-end encryption and DRM.

PATENTS & PUBLICATIONS

Aug. 2026. “Proofs Promptly: Proof-Oriented Programming with AI Agents (Experience Report)”. In: *Proceedings of the ACM International Conference on Functional Programming (ICFP)*. DOI: [10.1145/3828709](https://doi.org/10.1145/3828709).

June 2026. “Lazy Validation and Self-Healing for Agentic Programs”. In: *Proceedings of the ACM Workshop on Principles of Agentic Engineering (PAGe), Programming Language Design and Implementation (PLDI)*. Boulder, CO, USA. DOI: [10.1145/3819802.3820582](https://doi.org/10.1145/3819802.3820582).

- Oct. 2025.** “Structural temporal logic for mechanized program verification”. In: *Proceedings of the ACM International Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA)*.
- 🏆 **Distinguished Artifact Award.**
- Mar. 2025.** “Authorization Policy Validation”. US Patent 12,261,888.
- Dec. 2024.** “Choice Trees: Representing and reasoning about nondeterministic, recursive, and impure Programs in Coq”. In: *Journal of Functional Programming (JFP), Special POPL 2025 Edition*.
- Oct. 2024.** “Cedar: A New Language for Expressive, Fast, Safe, and Analyzable Authorization”. In: *Proceedings of the ACM International Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA)*.
- June 2024.** “Reef: Fast Succinct Non-Interactive Zero-Knowledge Regex Proofs”. In: *Proceedings of the USENIX Security Symposium (USENIX Security)*.
- June 2022.** “Efficient representation of numerical optimization problems for SNARKs”. In: *Proceedings of the USENIX Security Symposium (USENIX Security)*.
- Mar. 2020.** “Privacy-preserving system for machine-learning training data”. US Patent 10,601,786.
- Nov. 2019.** “Extracting and optimizing formally verified code for systems programming”. In: *Proceedings of the International Symposium on NASA Formal Methods (NFM)*.

COMMITTEE SERVICE

- 2026 PLDI Program Committee
- 2026 CSF Program Committee
- 2026 Rocqshop Program Committee